Jesse Taube

Embedded Software Engineer
Newton Massachusetts
University of Massachusetts Lowell

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GitHub Profile
LinkedIn Profile
My website

Technical Skills and Interests

Languages: C, C++, Bash, Verilog, Python, Rust
Tools & Technologies: Git, KVM, Podman, Linux Kernel, U-Boot, Buildroot, Debian
Protocols & Libraries: Networking, C&C++ STL, I^2C, SPI, RS232
Hardware: Experienced with reading schematics, using oscilloscopes, multi-meters, and low-speed routing
Areas of Interest: Embedded Linux, IOT and IOT security

Experience

- **Rivos Inc** — *May 2024 - Sep 2024, May 2025 - Aug 2025*
  Kernel Developer — Remote
  - RISC-V architecture improvements in Linux Kernel
  - Python and Bash based test fixtures
  - Custom Debugging hardware in QEMU and Riscv-isa-sim
  - Implementing SBI extentions in Rust
- **D3 Engineering** — *May 2023 - Aug 2023*
  Intern Kernel developer on the Nvidia Jetson Team — Rochester, New York
  - Linux Kernel driver development in C for GMSL and FPD-link cameras
  - Writing device trees and device tree overlays
  - Reading schematics and datasheets
  - Working with oscilloscopes and multi-meters
  - Debugging hardware and software issues, such as MIPI CSI data loss
- **Part-time Freelancer** — *Jun 2018 - Present*
  Hardware and Software developer — Hybrid
  - Writing Kernel drivers for custom hardware
  - Development of custom test boards
  - UL-certified SOM-based custom motor control board with HDMI
  - Programming tutor for University and high school students

Personal Projects

- **Linux Ports**
  Porting of IMXRT and RP2350(Raspberry Pi Pico 2) to the Linux Kernel
  - Writing Device Trees and Kernel drivers in C
  - Debugging over JTAG with GDB
  - Using agile development with co-developers
- **KISC-V**
  A KISS principle RISC-V32I CPU, written in SystemVerilog
  - Written in SystemVerilog and debugged with GTKWave
  - Used Vidado and Quartus to run on Xilinx and Altera FPGAs
  - Microcode generation in Python
  - Custom trap handler in C and Assembly to emulate CSR instructions
  - Can run No-MMU Linux
- **Personal Blog**
  Blog with backend in NodeJS hosted Locally
  - Dynamic and static pages in NodeJS using Express
  - Jquery and Bootstrap for frontend
  - Locally hosted on servers running Cockpit's KVM Dashboard

Education

- **University of Massachusetts Lowell** — *2022-26*
  Bachelor of Computer Engineering, Minor in Mathematics
- **Newton South High School** — *2022*
  - LigerBots - FRC team 2877
  - Science team